INF2001 Introduction to Software Engineering

Milestone 2: Detailed Design, Test Plan and Wireframe

Music School Workload Allocation & Resource Scheduling System

Prepared by

Team P5-5

Student Name	Student ID
Joseph Poon	2501079
Ryan Ke	2501911
Wei Xiang	2500686
Tan Yan Qi	2500030
Zayne Lee	2503844
Lab Group:	P5 - Team 5
Date:	20 November 2026

Table of Contents

Non-Plagiarism Declaration **3**

AI Usage **4**

1 Introduction **5**

1.1 Team member contributions 5

1.2 Final requirements 5

1.3 Final Class Diagram 9

2 Component Diagram **10**

2.1 Component Architecture & Subsystems 10

2.2 Provided and Required Interface Contracts 11

2.3 Architectural Design Rationale 12

3 Detailed Design **13**

3.1 Architecture Design 13

3.2 Detailed Design (Strategy & Observer Patterns) 14

4 Testing **18**

4.1 White Box Testing (Basis Path Coverage) 18

4.2 Black Box Testing (Decision Table & Boundary Parameters) 22

5 Wireframe **27**

5.1 Manager Scheduling & Outlier Intelligence View 27

5.2 Allocation Matrix & Conflict Feedback View 29

5.3 Teacher Availability & Timetable Inspection View 31

5.4 Audit Log Console, Admin Configuration & Design System View 33

6 References **35**

Non-Plagiarism Declaration

Students are not allowed to copy assessment solutions from another person or other places. It is the students' responsibility to guarantee that their solutions are their own work. Meanwhile, the students must also ensure that their work is not accessible by others. Where such plagiarism is detected, both of the assessments involved will receive ZERO mark.

We hereby declare that:

- We fully understand and agree to the abovementioned policy.
- We did not copy any materials from others or from other places.
- We did not share our materials with others or upload to any other places for public access.
- We agree that we will not disclose any information or material of the team project to others or upload to any other places for public access.
- We agree that our project will receive Zero mark if any misalignment with the abovementioned policies is detected.

Declared by: Group P5-5

Team members:

1. Joseph Poon (2501079)
2. Ryan Ke (2501911)
3. Wei Xiang (2500686)
4. Tan Yan Qi (2500030)
5. Zayne Lee (2503844)

Date: 20 November 2026

AI Usage

In accordance with SIT academic policy, generative AI tools were used as assistive engineering aids during Milestone 2 preparation, as summarized below:

Document Section	Generative AI Tool	Specific Prompt Intent & Usage Scope	Human Governance, Verification & Modifications
§1.2, §4.1 (Tables)	Claude Sonnet 5.5	Markdown table formatting, syntax linting, column alignment.	Replaced generic descriptions with concrete domain attributes; verified truth tables and algebraic reduction.
§1.3, §2.0, §3.2 (Diagrams)	Claude Opus 5.5	PlantUML syntax verification, styling parameter checks.	Manually authored all component boundaries, interface contracts, and GoF Strategy structures.
§2.3, §3.1 (Narrative)	Claude Sonnet 5.5	Phrasing refinement and technical editing.	Rewrote output to accurately reflect stakeholder interview evidence and statutory citations.
§4.1, §4.2 (Testing Suite)	Claude Opus 5.5	Control Flow Graph edge verification and cyclomatic complexity check.	Derived non-binary 3-tier workload basis paths manually, confirmed $V(G) = 6$, and defined full-coverage test parameter specifications.
§5.0 (Wireframes)	Figma + Figwright MCP	Programmatic Figma layout generation, design token binding, and high-DPI PNG asset export.	Verified static screenshot coverage across all required UI states and traced requirements (`FR-01` to `FR-18`, `NFR-USE-02`, `NFR-COM-01`).

Engineering Integrity Confirmation: All core business logic—including the 4-hour continuous teaching limit with mandatory 1-hour rest, instrument-equipped studio matching, monthly 19th 23:59 cutoff window, 2-stage rejection discussion lifecycle, and 3-tier workload classification (`< 40.0h` OK, `40.0h–44.0h` Warning, `> 44.0h` Overworked)—was derived directly from stakeholder interviews and manually verified against project specifications.

1 Introduction

1.1 Team member contributions

Student Name	Student ID	Project Role	Authored Sections & Responsibilities
Joseph Poon	2501079	Lead Architect	Authored the UML 2.5 Component Diagram (§2.0), formulated Service Interface Contracts (§2.2), restructured the Final Class Diagram (§1.3), and designed the White-Box Testing Control Flow Graph (§4.1).
Ryan Ke	2501911	Requirements & QA Lead	Addressed M1 consultation feedback, refined master Functional and Non-Functional Requirements (§1.2), developed Black-Box Decision Table testing methodology, and derived combinatorial reduction (§4.2).
Wei Xiang	2500686	Detailed Design Lead	Conducted Java / Spring Boot architectural pattern analysis (§3.1), evaluated GoF design pattern trade-offs (§3.2.1), formalized Strategy Pattern implementation for candidate tie-breaking (§3.2.2), and authored Observer pattern specifications (§3.2.3).
Tan Yan Qi	2500030	Test & Verification Lead	Formulated interface method contracts for inter-component communication (§2.2), validated non-binary 3-tier basis path coverage and cyclomatic complexity mathematics (§4.1), and specified full-coverage test case parameters.
Zayne Lee	2503844	Operations Lead	Developed architectural design rationale (§2.3), verified alignment between requirements, use cases, and static wireframe views (§5.0), and compiled the compliance audit trail specification.

1.2 Final requirements

Following the Milestone 1 consultation, the requirement baseline was refined to eliminate scope creep, resolve vague terminology, and enforce atomic specifications meeting the 7 traits of good requirements (cohesive, complete, consistent, atomic, traceable, unambiguous, verifiable).

The bilateral peer lesson swap feature (`FR-11` in M1) and intermediate draft schedule transition screens were removed from the active system scope to ensure sufficiency. Resolving lesson conflicts remains the operational responsibility of the School Manager through the job rejection discussion workflow, and confirming lesson allocations directly triggers teacher notifications. Because staff work allocation is assigned monthly, each slot carries a predetermined studio location that is displayed directly to teachers and managers. Teacher weekly workload is evaluated across three non-binary tiers: below 40.0 hours (`OK`), 40.0 to 44.0 hours (`Warning` / authorized overtime), and above 44.0 hours (`Overworked` / statutory ceiling breach). To ensure managers have sufficient scheduling flexibility, instructors must declare at least 60.0 hours of monthly availability.

Table 1.1: Final Functional Requirements Specification

ID	Priority	Functional Requirement Specification	Updates from M1
FR-01	Must	The system shall display each teacher's monthly work allocation and assigned studio location upon lesson confirmation.	Removed running weekly/monthly hours calculation; focused on displaying monthly allocation and assigned location.
FR-02	Must	The system shall sort and display the top 3 teachers with the lowest allocated hours on the manager dashboard and flag teachers across three workload tiers: below 40.0 hours (`OK`), 40.0 to 44.0 hours (`Warning`), and above 44.0 hours (`Overworked`).	Removed vague term "dynamically surface"; aligned with 3-tier non-binary workload model.
FR-03	Must	The system shall allow managers to manually allocate unassigned 30-minute lesson slots to eligible teachers for a target week.	
FR-04	Must	The system shall display up to 3 candidate teachers side-by-side during allocation, displaying each candidate's current workload, teaching preferences, and assigned studio location for the specified date.	Clarified that physical studio location is displayed from the slot posting.
FR-05	Should	The system shall rank eligible candidate teachers by remaining weekly capacity below the 40.0-hour threshold and preferred teaching day match.	Removed parenthetical formula verbosity; replaced vague "teaching conditions".
FR-06	Must	The system shall allow teachers to submit and edit recurring 30-minute availability slots up to 1 month ahead until 23:59 on the 19th of each month, enforcing a minimum declaration of 60.0 hours.	Added minimum 60.0-hour availability threshold.
FR-07	Must	The system shall allow teachers to specify preferred teaching days, preferred studio ID, and consecutive lesson block limits of up to 4.0 continuous hours followed by a minimum 1.0-hour rest break within their submitted availability window.	Clarified relationship between availability and preferences; defined consecutive block rules.
FR-08	Must	The system shall provide teachers with a personal weekly schedule displaying confirmed lessons, assigned studio locations, and enrolled student details.	Replaced over-specific masked contact phrasing with enrolled student details.
FR-09	Must	The system shall allow teachers to reject an assigned lesson slot by recording a rejection request flagged for discussion with the manager prior to re-allocation.	Simplified rejection workflow wording to reflect manager consultation before proceeding.
FR-10	Should	The system shall flag availability changes submitted after the 19th cutoff for manager review.	Simplified late availability review phrasing.
FR-11	Should	The system shall allow IT administrators to schedule studio maintenance periods and exclude studios under maintenance from lesson allocation.	Simplified maintenance blackout terminology to studio maintenance periods.

FR-12	Should	The system shall allow managers to reassign scheduled lessons of an absent teacher to qualified and available substitute teachers.	Generalized from batch reassignment to single-slot substitute reassignment.
FR-13	Must	The system shall allow IT administrators to provision user accounts and perform logical account deactivations for Teachers, Managers, and IT Administrators.	
FR-14	Must	The system shall allow IT administrators to configure studio instrument equipment for Studios 1 to 5 and restrict lesson allocations to studios equipped with the required instrument.	Generalized Studio 5 hardcoding to instrument-equipped studio matching.
FR-15	Must	The system shall notify assigned teachers when lesson allocations are confirmed for a weekly schedule.	Removed intermediate draft release schedule transitions while preserving teacher notification.
FR-16	Should	The system shall calculate booked versus available studio hours across the 86 weekly operating hours and generate studio utilisation reports for school administration.	Standardized British spelling (`utilisation`) and retained as `Should` priority for administration completeness.
FR-17	Must	The system shall evaluate teacher weekly working hours across three tiers—acceptable (`< 40.0` hours), overtime warning (`40.0–44.0` hours), and overworked statutory breach (`> 44.0` hours under Employment Act Part IV)—and require manager confirmation for allocations in the warning tier.	Unified statutory and overtime checks into the 3-tier non-binary workload model.
FR-18	Must	The system shall append an immutable audit log record for every lesson allocation, overtime confirmation, rejection update, and administrative configuration change, and allow managers to inspect the audit trail.	Retained immutable audit logging and inspection capability.

Table 1.2: Final Non-Functional Requirements Specification

NFR ID	Category	Non-Functional Requirement Specification	Operational & Statutory Justification
NFR-USE-01	Usability	The system shall render all web interfaces across desktop and mobile viewports (375px to 1920px) without horizontal scrolling.	Enables mobile timetable inspection for teachers and desktop allocation for managers.
NFR-USE-02	Usability	The system shall display teacher weekly workload status using three visual tiers paired with text badges: Green `OK` (`< 40.0h`), Amber `Warning` (`40.0h–44.0h`), and Red `Overworked` (`> 44.0h`).	Aligns UI visual indicators with the 3-tier workload classification and WCAG 2.2 AA accessibility rules.

NFR-PER-01	Performance	The system shall complete initial page rendering within 1.5 seconds under a concurrent load of 20 active users.	Ensures responsive scheduling operations during monthly allocation cycles.
NFR-PER-02	Performance	The system shall evaluate candidate eligibility and clash constraints within 300 milliseconds of slot selection.	Delivers immediate decision support during manual lesson slot allocation.
NFR-REL-01	Reliability	The system shall maintain at least 99.5% uptime during the monthly scheduling window from the 19th to the 24th of each calendar month.	Prevents scheduling bottlenecks around the monthly availability cutoff.
NFR-REL-02	Reliability	The system shall execute account and lesson removals exclusively via logical status flags without physical record deletion.	Preserves historical referential integrity for audit and utilisation reporting.
NFR-SEC-01	Security	The system shall enforce role-based access control restricting teachers to their own schedules and restricting system configuration panels to IT administrators.	Protects staff schedules and prevents unauthorized administrative changes.
NFR-COM-01	Compliance	The system shall retain immutable audit log records for at least 24 calendar months.	Satisfies Ministry of Manpower statutory employment record retention requirements.
NFR-COM-02	Compliance	The system shall enforce the 44.0-hour weekly maximum under Employment Act Part IV and record all authorized overtime allocations between 40.0 and 44.0 hours.	Ensures statutory labor compliance across all teaching assignments.

1.3 Final Class Diagram

The final class diagram incorporates all structural updates from Milestone 1 and Milestone 2, including explicit interface realizations, Java / Spring Boot service contracts, teacher workload and assigned location attributes, and design pattern roles.

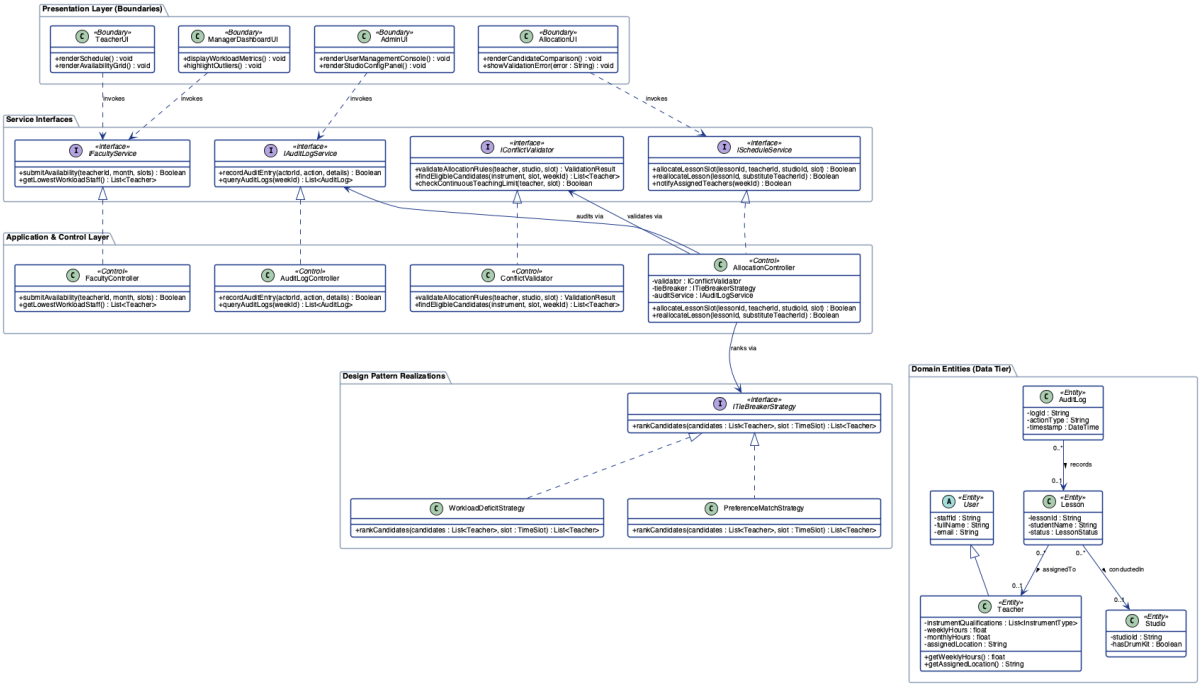


Figure 1.1: Final Class Diagram with Service Interfaces, Controllers, Strategies, and Domain Entities

2 Component Diagram

2.1 Component Architecture & Subsystems

The component diagram partitions the **Java 21 and Spring Boot** application into three primary internal layers enclosed within the system boundary, exposing clean Spring service interfaces and isolating external infrastructure gateways.

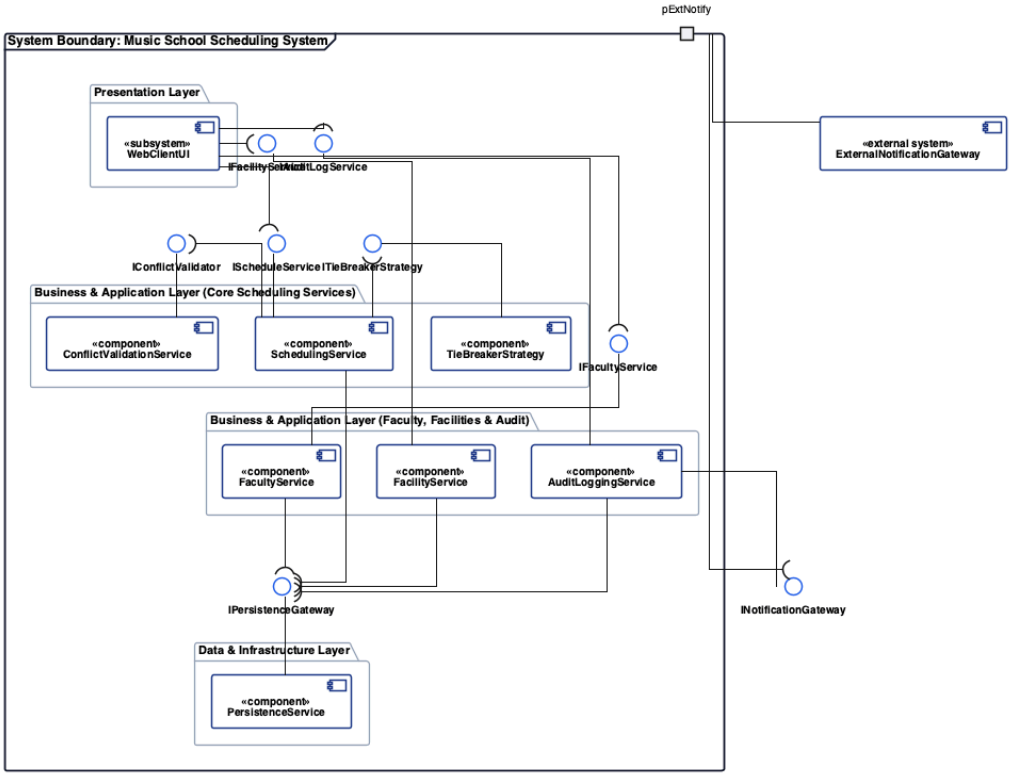


Figure 2.1: UML 2.5 Component Diagram for Music School Scheduling System

The system architecture comprises seven primary components implemented as modular Spring Boot components:

1. WebClientUI: Presentation subsystem encapsulating boundary controllers (`AllocationUI`, `ManagerDashboardUI`, `TeacherUI`, `AdminUI`, `AuditLogUI`). Operates strictly as a service consumer invoking provided Java service interfaces.
2. SchedulingService: Central Spring `@Service` application controller orchestrating lesson slot allocation, substitute reassignment, rejection review, and teacher schedule notification.
3. ConflictValidationService: Domain validation component executing multi-constraint checks across instrument certifications, room instrument compatibility, 4.0-hour continuous teaching limits, and the 3-tier weekly workload model (`< 40.0h`, `40.0h–44.0h`, `> 44.0h`).
4. FacultyService: Manages teacher profiles, assigned studio locations, availability grids, monthly submission windows, and workload calculations.
5. FacilityService: Manages physical room metadata for Studios 1 to 5, studio instrument configurations, studio utilisation metrics, and scheduled studio maintenance periods.
6. AuditLoggingService: Centralized regulatory logging service maintaining append-only historical ledgers and compliance query tools via Spring Data JPA.
7. ExternalNotificationGateway: Infrastructure adapter situated outside the system boundary, interfacing via boundary port `pExtNotify` to dispatch automated timetable alerts to assigned

teachers.

2.2 Provided and Required Interface Contracts

Every exposed interface realizes strict method contracts ensuring compile-time safety and decoupled communication across component boundaries:

`IScheduleService` (Provided by SchedulingService, Required by WebClientUI):

- `allocateLessonSlot(lessonId: String, teacherId: String, studioId: String, slot: TimeSlot): Boolean`
- `reallocateLesson(lessonId: String, substituteTeacherId: String): Boolean`
- `notifyAssignedTeachers(weekId: String): Boolean`

`IConflictValidator` (Provided by ConflictValidationService, Required by SchedulingService):

- `validateAllocationRules(teacher: Teacher, studio: Studio, slot: TimeSlot): ValidationResult`
- `findEligibleCandidates(instrument: InstrumentType, slot: TimeSlot, weekId: String): List<Teacher>`
- `checkContinuousTeachingLimit(teacher: Teacher, slot: TimeSlot): Boolean`

`IFacultyService` (Provided by FacultyService, Required by WebClientUI, SchedulingService, ConflictValidationService):

- `submitAvailability(teacherId: String, month: Date, slots: List<TimeSlot>): Boolean`
- `getLowestWorkloadStaff(): List<Teacher>`
- `getTeacher(teacherId: String): Teacher`

`IFacilityService` (Provided by FacilityService, Required by WebClientUI, ConflictValidationService):

- `configureStudio(studioId: String, name: String, hasPiano: Boolean): Boolean`
- `flagMaintenanceWindow(studioId: String, start: DateTime, end: DateTime, reason: String): Boolean`
- `getStudio(studioId: String): Studio`

`IAuditLogService` (Provided by AuditLoggingService, Required by WebClientUI, SchedulingService):

- `recordAuditEntry(actorId: String, action: String, details: String): Boolean`
- `queryAuditLogs(weekId: String): List<AuditLog>`

`ITieBreakerStrategy` (Provided by TieBreakerStrategy, Required by SchedulingService):

- `rankCandidates(candidates: List<Teacher>, slot: TimeSlot): List<Teacher>`

2.3 Architectural Design Rationale

The software architecture prioritizes three critical quality attributes derived from stakeholder elicitation:

1. Modifiability: Business rules for music academies evolve frequently, including shifting statutory overtime caps, new instrument certifications, and updated break policies. Decoupling validation logic into `ConflictValidationService` and ranking algorithms into `ITieBreakerStrategy` allows developers to modify business rules without altering presentation components or data storage models.
2. Reliability & Data Integrity: Manual spreadsheet allocation resulted in undetected room double-bookings and unmonitored overtime. Centralizing placement through `IScheduleService` guarantees that every assignment passes mandatory conflict detection before persistence. Encapsulating mutation logs in `IAuditLogService` guarantees a 24-month immutable compliance history satisfying Ministry of Manpower mandates.
3. Usability & Operational Responsiveness: School managers construct complex weekly schedules under strict deadlines. Decomposing the system into granular service interfaces enables sub-300ms conflict checks on candidate selection, giving managers immediate visual feedback without reloading full page states.

3 Detailed Design

3.1 Architecture Design

The system implements a strict Hierarchical Layered Architecture in **Java 21 with Spring Boot**, organized into three closed tiers: Presentation Layer (Spring MVC / Thymeleaf Web Boundary), Business and Application Layer (Spring `@Service` Domain Controllers and Strategies), and Data and Infrastructure Layer (Spring Data JPA Repositories with PostgreSQL and SMTP/SMS Gateways).

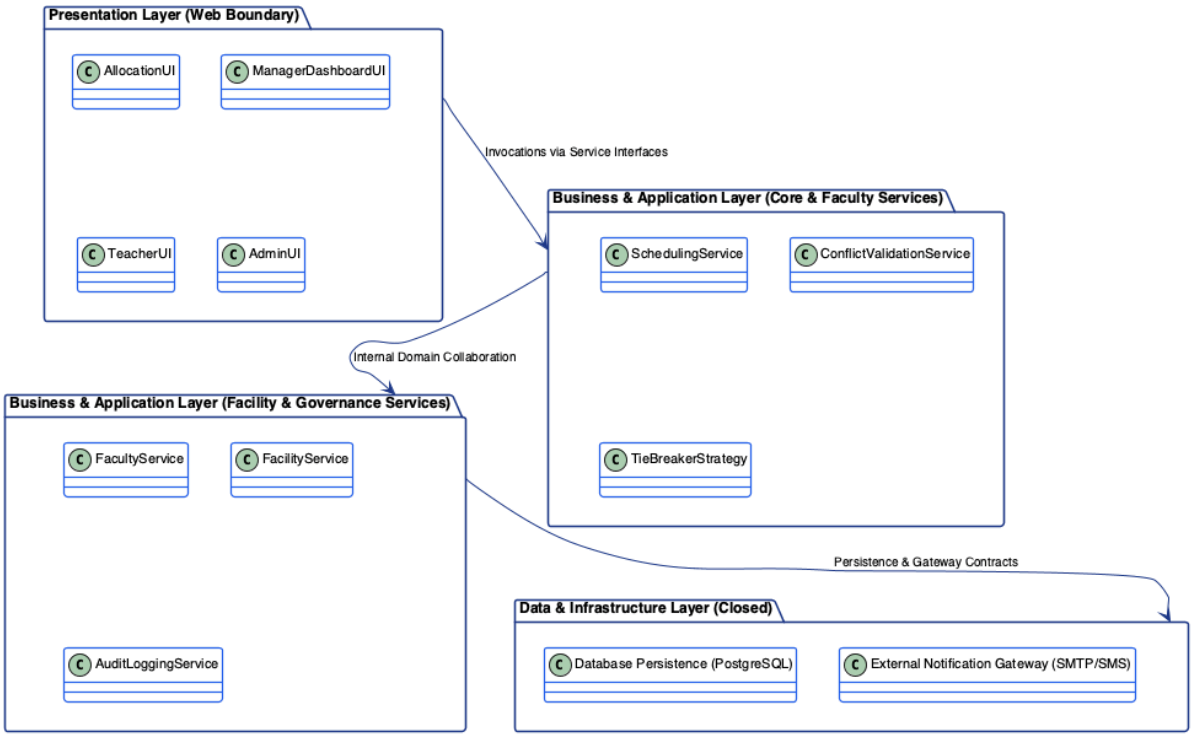


Figure 3.1: 3-Tier Hierarchical Layered Architecture for Music School Scheduling System

In accordance with closed layering rules, layer-skipping is strictly prohibited. Requests flow sequentially downward through the business controllers, and domain data returns synchronously back through the calling hierarchy. Architectural Pattern Comparison:

Evaluated Architecture	Structural Characteristics	Suitability Analysis for Scheduling System
Hierarchical Layered (Selected)	Partitions system horizontally into closed Presentation, Business, and Persistence layers. Each layer depends only on the layer immediately below it.	Selected: Enforces clean separation of concerns, guarantees that business rules and statutory overtime checks cannot be bypassed by UI components, and simplifies unit test isolation.
Model-View-Controller (MVC)	Separates UI rendering (View) and user inputs (Controller) from domain entities (Model).	Adopted within Presentation Tier: Boundary views capture user inputs and delegate to controllers, but MVC alone lacks formal boundaries for complex domain validation services.

Microservices Architecture	Distributes application logic across independent, network-partitioned service processes.	Rejected: Introduces unnecessary network latency, distributed transaction complexity, and DevOps overhead for an academy with 9 teachers and 5 studios.

3.2 Detailed Design

3.2.1 Design Pattern Evaluation & Trade-off Analysis

To resolve detailed design problems within the scheduling core, four Gang of Four (GoF) patterns were evaluated against operational requirements:

Evaluated Pattern	Target Operational Problem	Architectural Trade-offs & Analysis	Selection Outcome
Strategy Pattern	Dynamic candidate teacher ranking across competing operational heuristics (workload deficit vs preference match).	Pros: Decouples candidate sorting algorithms from allocation controllers; satisfies Open/Closed Principle. Cons: Increases class count slightly. Verdict: Long-term extensibility outweighs minimal class boilerplate.	Selected (Implemented)
Observer Pattern	Decoupling schedule publication events from automated email, SMS, and dashboard notification dispatch.	Pros: Broadcasts schedule publication events to multiple communication channels without coupling scheduling controllers to notification protocols. Cons: Unsubscribing listeners requires explicit lifecycle management to avoid memory retention.	Selected (Implemented)
Factory Method	Instantiating specialized User accounts (Teacher, Manager, ITAdministrator) during onboarding.	Pros: Encapsulates role creation. Cons: Account onboarding occurs infrequently; adds unnecessary abstraction over simple constructors.	Evaluated (Alternative)
Singleton Pattern	Providing global access to the scheduling engine or system configuration registry.	Pros: Single point of access. Cons: Introduces global state, tightly couples callers to concrete implementations, and hinders test isolation.	Evaluated (Rejected)

3.2.2 Strategy Pattern (Selected)

Operational Problem: When allocating an unassigned lesson, multiple teachers often satisfy all hard eligibility constraints (qualified, available, within 4.0-hour continuous teaching limits, and at or below the 44.0-hour cap). The system must rank these candidates to guide manager selection. Under nominal operations, management prioritizes faculty equity by ranking teachers by remaining capacity below the 40.0-hour threshold. In specialized cases, management prioritizes teacher satisfaction by ranking candidates by preferred teaching days and studio assignments.

Hardcoding this ranking logic into `AllocationController` violates the Open/Closed Principle (OCP) and Single Responsibility Principle (SRP). Adding new ranking heuristics requires modifying and recompiling core scheduling classes. The Strategy Pattern decouples algorithmic ranking from scheduling orchestration.

Structural Template:

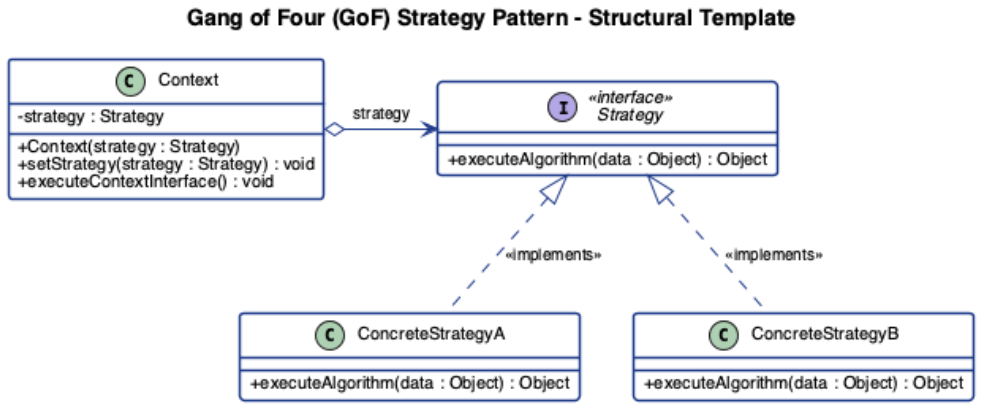


Figure 3.2: Gang of Four Strategy Pattern Standard Structural Template

Before Pattern Implementation:

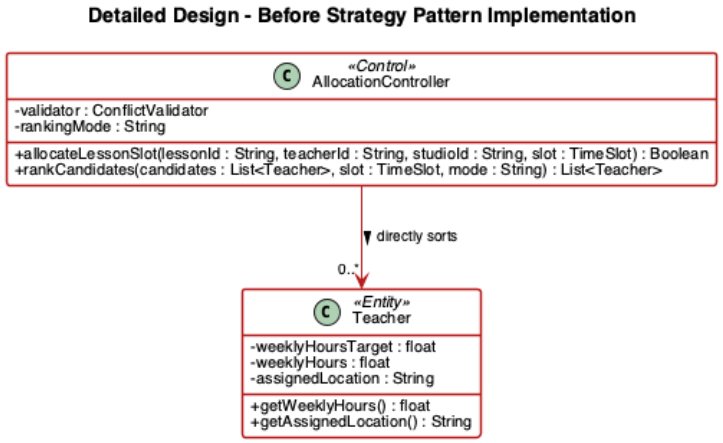


Figure 3.3: Class Excerpt Before Strategy Implementation (Monolithic Conditional Sorting)

After Strategy Pattern Implementation:

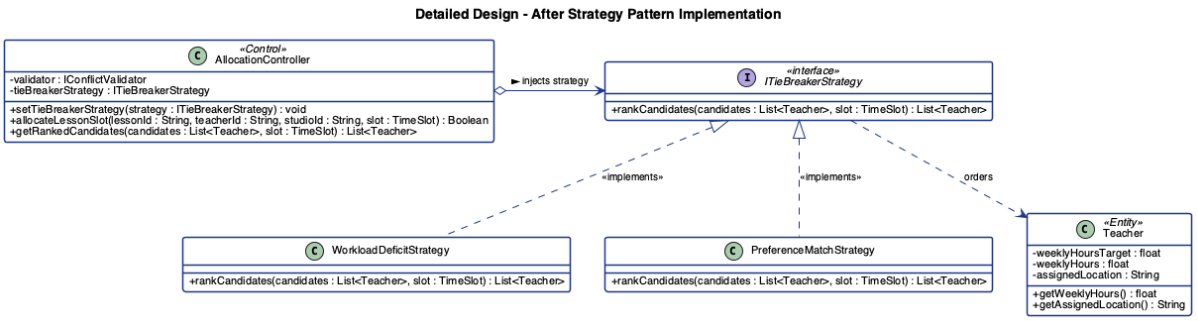


Figure 3.4: Class Excerpt After Strategy Implementation (Extensible Policy Decoupling)

In this refactored structure, `AllocationController` acts as the Context, holding a reference to the `ITieBreakerStrategy` interface. Concrete strategies `WorkloadDeficitStrategy` and `PreferenceMatchStrategy` encapsulate independent sorting algorithms. When managers toggle ranking preferences in `AllocationUI`, the controller dynamically swaps the active strategy instance without modifying orchestration code.

3.2.3 Observer Pattern (Selected)

Operational Problem: When a manager confirms weekly lesson allocations and notifies assigned teachers (`UC-05` / `FR-15`), multiple decoupled subsystems require notification: assigned teachers must receive schedule alerts, administrators require audit event records, and dashboard metrics must synchronize. Direct calls from `AllocationController` to external mail or SMS APIs tightly couple core scheduling logic to communication protocols.

Applying the Observer Pattern establishes a one-to-many dependency, allowing `AllocationController` (Subject) to broadcast state transitions to registered listeners (Observers) without coupling to concrete notification implementations.

Structural Template:

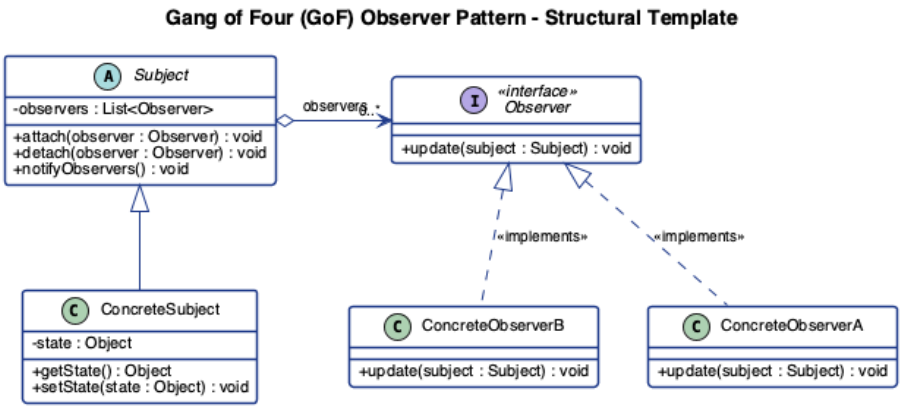


Figure 3.5: Gang of Four Observer Pattern Standard Structural Template

After Observer Pattern Implementation:

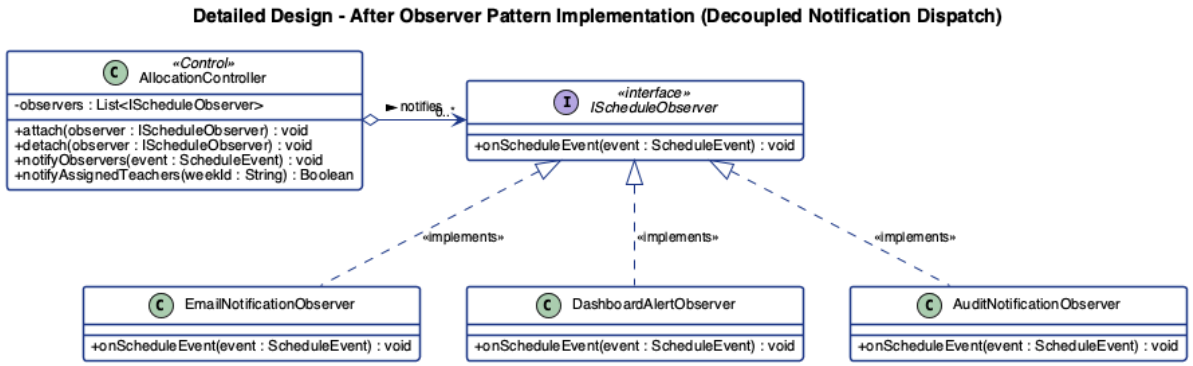


Figure 3.6: Class Excerpt After Observer Implementation (Decoupled Notification Dispatch)

In this structure, `AllocationController` maintains an internal subscriber list through the `IScheduleObserver` interface. When `notifyAssignedTeachers()` executes, it iterates through observers invoking `onScheduleEvent()`. Concrete subscribers `EmailNotificationObserver`, `DashboardAlertObserver`, and `AuditNotificationObserver` handle delivery independently.

4 Testing

4.1 White Box Testing (Basis Path Coverage)

White-box testing is conducted first using **Basis Path Testing** on `ConflictValidator.findEligibleCandidates(instrument, slot, weekId)` to guarantee 100% statement coverage and 100% branch coverage across all linearly independent execution paths.

4.1.1 Control Flow Graph (CFG) & Non-Binary 3-Tier Workload Classification

Rather than splitting workload evaluation into chained binary boolean flags, Node 5 (`N5`) models the teacher workload evaluation as a single **non-binary 3-way decision branch** corresponding directly to the three operational workload tiers:

- **Tier 1 (`< 40.0h` — OK):** Projected weekly hours remain below 40.0 hours (`N6`); candidate is added with status `OK`.
- **Tier 2 (`40.0h–44.0h` — Warning):** Projected weekly hours fall between 40.0 and 44.0 hours inclusive (`N7`); candidate is added with status `WARNING` requiring manager overtime confirmation.
- **Tier 3 (`> 44.0h` — Overworked):** Projected weekly hours exceed the 44.0-hour statutory ceiling (`N8`); candidate is marked `OVERWORKED` and excluded from the eligible list.

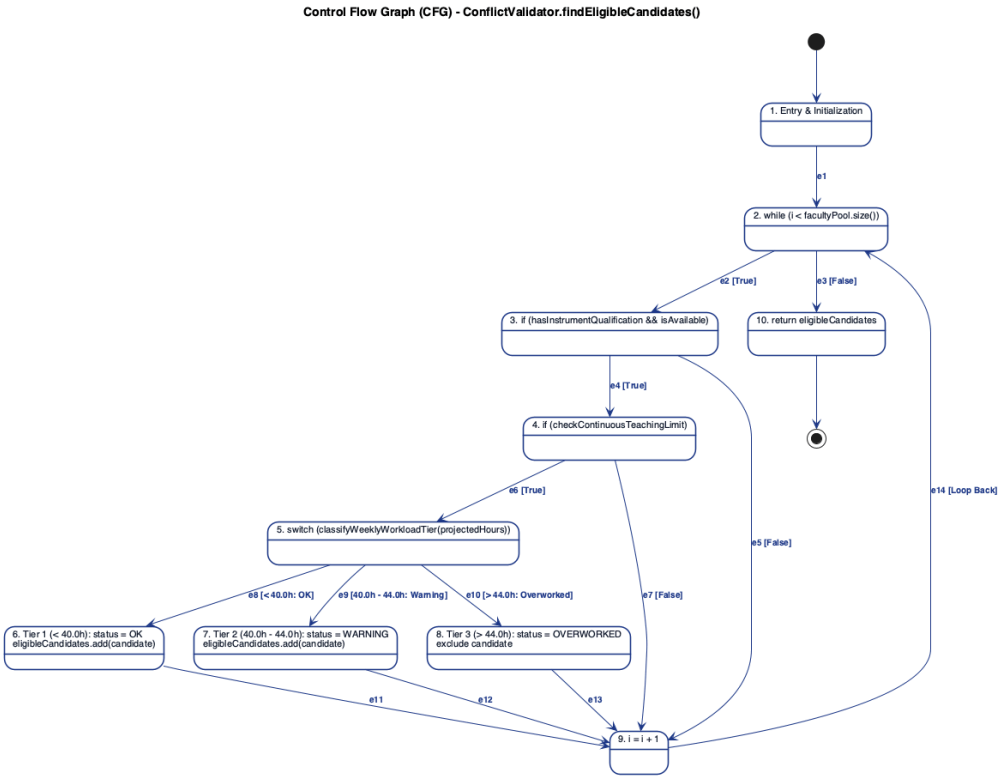


Figure 4.1: Control Flow Graph for ConflictValidator.findEligibleCandidates() with 3-Way Non-Binary Workload Branching

4.1.2 Cyclomatic Complexity Calculation

Cyclomatic complexity $V(G)$ is calculated using the McCabe graph-theoretic formula:

$$V(G) = E - N + 2P$$

- **Number of directed edges (E):** 14 edges (`e1` through `e14`)
- **Number of CFG nodes (N):** 10 nodes (`N1` through `N10`)
- **Number of connected components (P):** 1 connected method flow

$$V(G) = 14 - 10 + 2(1) = 6$$

Equivalently, counting the branching contribution of each predicate node ($\sum(\text{out-degree} - 1) + 1$): binary decisions `N2`, `N3`, and `N4` each contribute $2 - 1 = 1$, while the 3-way workload decision node `N5` contributes $3 - 1 = 2$, giving $V(G) = (1 + 1 + 1 + 2) + 1 = 6$. Therefore, exactly **6 linearly independent basis paths** are required to achieve complete path coverage of the basis set.

4.1.3 Linearly Independent Basis Paths

1. **Path 1 (Empty Faculty Pool):** `N1 -> N2 -> N10` (Edge sequence: `e1, e3`) — Loop condition `i < facultyPool.size()` evaluates `False` immediately.
2. **Path 2 (Unqualified or Unavailable Candidate):** `N1 -> N2 -> N3 -> N9 -> N2 -> N10` (Edge sequence: `e1, e2, e5, e14, e3`) — Candidate lacks required instrument qualification or slot availability at `N3`.
3. **Path 3 (Continuous Teaching Limit Exceeded):** `N1 -> N2 -> N3 -> N4 -> N9 -> N2 -> N10` (Edge sequence: `e1, e2, e4, e7, e14, e3`) — Candidate is qualified and available, but assigning the 0.5h slot exceeds the 4.0-hour continuous teaching limit at `N4`.
4. **Path 4 (Workload Tier 1 — Acceptable `< 40.0h`):** `N1 -> N2 -> N3 -> N4 -> N5 -> N6 -> N9 -> N2 -> N10` (Edge sequence: `e1, e2, e4, e6, e8, e11, e14, e3`) — Candidate passes `N3` and `N4`, and projected weekly hours `< 40.0h` route through `e8` to `N6` (`status = OK`, added to `eligibleCandidates`).
5. **Path 5 (Workload Tier 2 — Warning `40.0h–44.0h`):** `N1 -> N2 -> N3 -> N4 -> N5 -> N7 -> N9 -> N2 -> N10` (Edge sequence: `e1, e2, e4, e6, e9, e12, e14, e3`) — Candidate passes `N3` and `N4`, and projected weekly hours in `[40.0h, 44.0h]` route through `e9` to `N7` (`status = WARNING`, added to `eligibleCandidates` with overtime warning).
6. **Path 6 (Workload Tier 3 — Overworked `> 44.0h`):** `N1 -> N2 -> N3 -> N4 -> N5 -> N8 -> N9 -> N2 -> N10` (Edge sequence: `e1, e2, e4, e6, e10, e13, e14, e3`) — Candidate passes `N3` and `N4`, and projected weekly hours `> 44.0h` route through `e10` to `N8` (`status = OVERWORKED`, excluded from `eligibleCandidates`).

4.1.4 Full-Coverage White-Box Test Case Parameter Specifications

To ensure full statement, branch, basis-path, and boundary coverage across the 3-tier workload model (`< 40.0h`, `40.0h`, `40.0h–44.0h`, `44.0h`, `> 44.0h`) without redundant unit test pseudocode, Table 4.1 defines the exact input parameter vectors and expected outcomes for every basis path and tier boundary.

Table 4.1: White-Box Basis Path & Boundary Test Parameters for `findEligibleCandidates()`

Test ID	Basis Path	`facultyPool` Input Parameters (`staffId`, `quals`, `avail`, `contHours`, `weeklyHours`)	Query Parameters (`instrument`, `slot`, `weekId`)	Projected Hours (`weeklyHours + 0.5h`)	CFG Branch Taken	Expected Output & Coverage Role

TC-WB-01	Path 1	`[]` (Empty list, `size = 0`)	`instrument = PIANO`, `slot = Tue 14:00–14:30`, `weekId = 2026-W47`	N/A	`N2 -> N10` (`e3`)	Returns `[]` (size 0). Covers empty loop bypass.
TC-WB-02A	Path 2	`[Teacher("T-01", [VIOLIN], avail=true, contHours=1.0, weeklyHours=20.0)]`	`instrument = PIANO`, `slot = Tue 14:00–14:30`, `weekId = 2026-W47`	`20.5h`	`N3 -> N9` (`e5`)	Returns `[]` (size 0). Covers qualification failure in `N3`.
TC-WB-02B	Path 2	`[Teacher("T-02", [PIANO], avail=false, contHours=1.0, weeklyHours=20.0)]`	`instrument = PIANO`, `slot = Tue 14:00–14:30`, `weekId = 2026-W47`	`20.5h`	`N3 -> N9` (`e5`)	Returns `[]` (size 0). Covers availability failure in `N3`.
TC-WB-03	Path 3	`[Teacher("T-03", [PIANO], avail=true, contHours=4.0, weeklyHours=25.0)]`	`instrument = PIANO`, `slot = Tue 14:00–14:30`, `weekId = 2026-W47`	`25.5h`	`N4 -> N9` (`e7`)	Returns `[]` (size 0). Covers 4.0h continuous block boundary (`4.0 + 0.5 = 4.5 > 4.0`).
TC-WB-04	Path 4 (Tier 1)	`[Teacher("T-04", [PIANO], avail=true, contHours=3.5, weeklyHours=39.0)]`	`instrument = PIANO`, `slot = Tue 14:00–14:30`, `weekId = 2026-W47`	`39.5h` (`< 40.0h`)	`N5 -> N6` (`e8`)	Returns `["T-04"]` with `status = OK`. Covers Tier 1 upper boundary (`39.5h < 40.0h`) and `contHours = 3.5h` (`4.0h` valid boundary).

TC-WB-05A	Path 5 (Tier 2 Lower)	`[Teacher("T-05", [PIANO], avail=true, contHours=2.0, weeklyHours=39.5)]`	`instrument = PIANO`, `slot = Tue 14:00–14:30`, `weekId = 2026-W47`	`40.0h` (`40.0h–44.0h`)	`N5 -> N7` (`e9`)	Returns `["T-05"]` with `status = WARNING`. Covers exact `40.0h` warning tier entry boundary.
TC-WB-05B	Path 5 (Tier 2 Upper)	`[Teacher("T-06", [PIANO], avail=true, contHours=1.5, weeklyHours=43.5)]`	`instrument = PIANO`, `slot = Tue 14:00–14:30`, `weekId = 2026-W47`	`44.0h` (`40.0h–44.0h`)	`N5 -> N7` (`e9`)	Returns `["T-06"]` with `status = WARNING`. Covers exact `44.0h` statutory ceiling boundary.
TC-WB-06	Path 6 (Tier 3)	`[Teacher("T-07", [PIANO], avail=true, contHours=1.0, weeklyHours=44.0)]`	`instrument = PIANO`, `slot = Tue 14:00–14:30`, `weekId = 2026-W47`	`44.5h` (`> 44.0h`)	`N5 -> N8` (`e10`)	Returns `[]` (size 0), `status = OVERWORKED`. Covers `> 44.0h` overworked rejection branch.

4.2 Black Box Testing (Decision Table & Boundary Parameters)

Black-box testing complements the white-box basis path analysis by validating the external contract of `SchedulingService.allocateLessonSlot(lessonId, teacherId, studioId, slot)` using Equivalence Partitioning, Decision Table Analysis, and Boundary Value Analysis (BVA).

4.2.1 Cause-Effect Specification

The allocation contract evaluates five input causes where `C4` is a **3-valued non-binary condition** representing the projected weekly workload tier (`< 40.0h`, `40.0h–44.0h`, `> 44.0h`):

- **Causes (Inputs and Pre-conditions):**
  - `C1 {qual}`: Teacher holds certified instrument qualification for the lesson (`True` / `False`)
  - `C2 {avail}`: Teacher declared available for the selected 30-minute timeslot (`True` / `False`)
  - `C3 {cont}`: Teacher continuous teaching block with this slot is ≤ 4.0 hours (`True` / `False`)
  - `C4 {workloadTier}`: Projected weekly hours (`weeklyHours + 0.5h`) classification:
    - `OK`: `< 40.0` hours
    - `WARNING`: `40.0` to `44.0` hours inclusive
    - `OVERWORKED`: `> 44.0` hours
  - `C5 {studio}`: Assigned studio is unoccupied, not in a scheduled studio maintenance period (`FR-11`), and equipped with the required instrument (`FR-14`) (`True` / `False`)
- **Effects (System Actions and Outputs):**
  - `E1 {confirm_ok}`: Lesson confirmed (`OK`); assigned teacher notified (`FR-15`) and audit record appended (`FR-18`).
  - `E2 {confirm_warn}`: Overtime warning triggered (`40.0h–44.0h`); confirms upon manager authorization and logs overtime audit entry (`FR-17`, `FR-18`).
  - `E3 {qual_err}`: Allocation rejected: `"Teacher not qualified for instrument"`.
  - `E4 {avail_err}`: Allocation rejected: `"Teacher unavailable for selected timeslot"`.
  - `E5 {cont_err}`: Allocation rejected: `"Continuous 4.0h teaching limit exceeded (1.0h rest required)"`.
  - `E6 {overworked_err}`: Allocation rejected: `"Teacher overworked (> 44.0h statutory weekly limit breached)"`.
  - `E7 {studio_err}`: Allocation rejected: `"Studio unavailable, under maintenance, or lacks required instrument equipment"`.

4.2.2 Complete & Reduced Decision Table

With four binary conditions (`C1`, `C2`, `C3`, `C5`) and one 3-valued condition (`C4`), the complete combination space contains $2^4 \times 3 = 48$ rules. Collapsing don't-care states (`-`) where earlier constraint failures short-circuit downstream checks reduces the space to **7 canonical decision rules**:

Table 4.2: Reduced Decision Table for `allocateLessonSlot()`

Conditions / Rules	R1	R2	R3	R4	R5	R6	R7
C1 {qual}	**F**	**T**	**T**	**T**	**T**	**T**	**T**

C2 {avail}	-	**F**	**T**	**T**	**T**	**T**	**T**
C3 {cont} ($\leq 4.0\text{h}$)	-	-	**F**	**T**	**T**	**T**	**T**
C4 {workloadTier}	-	-	-	**OVERWORKED (>** `44.0h`**)**	`OK` / `WARNING`	**OK (<** `40.0h`**)**	**WARNING (**`40.0h`**–** `44.0h`**)**
C5 {studio}	-	-	-	-	**F**	**T**	**T**
Effects							
E1 {confirm_ok}						**X**	
E2 {confirm_warn}							**X**
E3 {qual_err}	**X**						
E4 {avail_err}		**X**					
E5 {cont_err}			**X**				
E6 {overworked_err}				**X**			
E7 {studio_err}					**X**		

4.2.3 Black-Box Test Case Parameter Matrix (Equivalence Partitions & BVA)

Table 4.3 specifies the concrete parameter values covering all 7 decision rules and every boundary point across the continuous teaching block (`3.5h`, `4.0h`) and the 3-tier weekly workload thresholds (`39.0h -> 39.5h`, `39.5h -> 40.0h`, `43.5h -> 44.0h`, `44.0h -> 44.5h`).

Table 4.3: Black-Box Test Cases & Boundary Parameters for `allocateLessonSlot()`

Test ID	Rule	Input Parameters (`lessonId`, `instrument`, `teacherId`, `quals`, `avail`, `contHours`, `weeklyHours`, `studioId`, `studioState`)	Projected Weekly Hours	Expected System Outcome
TC-BB-01	R1	`lessonId="L-101"`, `instrument=DRUMS`, `teacherId="T-01"`, `quals=[PIANO]`, `avail=true`, `contHours=1.0h`, `weeklyHours=22.0h`, `studioId="S-05"`, `studioState=FREE_DRUM_EQUIPPED`	`22.5h` (`OK`)	Rejected with `E3 {qual_err}`.
TC-BB-02	R2	`lessonId="L-102"`, `instrument=PIANO`, `teacherId="T-02"`, `quals=[PIANO]`, `avail=false`, `contHours=1.0h`, `weeklyHours=22.0h`, `studioId="S-01"`, `studioState=FREE_PIANO_EQUIPPED`	`22.5h` (`OK`)	Rejected with `E4 {avail_err}`.

TC-BB-03	R3	`lessonId="L-103"`, `instrument=PIANO`, `teacherId="T-03"`, `quals=[PIANO]`, `avail=true`, `contHours=4.0h`, `weeklyHours=28.0h`, `studioId="S-01"`, `studioState=FREE_PIANO_EQUIPPED`	`28.5h` (`OK`)	Rejected with `E5` `{cont_err}` (continuous block reaches `4.5h > 4.0h`).
TC-BB-04	R4	`lessonId="L-104"`, `instrument=PIANO`, `teacherId="T-04"`, `quals=[PIANO]`, `avail=true`, `contHours=1.5h`, `weeklyHours=44.0h`, `studioId="S-01"`, `studioState=FREE_PIANO_EQUIPPED`	`44.5h` (`OVERWORKED`)	Rejected with `E6` `{overworked_err}` (exceeds `44.0h` statutory cap).
TC-BB-05A	R5	`lessonId="L-105"`, `instrument=PIANO`, `teacherId="T-05"`, `quals=[PIANO]`, `avail=true`, `contHours=1.0h`, `weeklyHours=30.0h`, `studioId="S-03"`, `studioState=MAINTENANCE_PERIOD`	`30.5h` (`OK`)	Rejected with `E7` `{studio_err}` (studio in scheduled maintenance period per `FR-11`).
TC-BB-05B	R5	`lessonId="L-106"`, `instrument=DRUMS`, `teacherId="T-06"`, `quals=[DRUMS]`, `avail=true`, `contHours=1.0h`, `weeklyHours=30.0h`, `studioId="S-01"`, `studioState=FREE_PIANO_ONLY`	`30.5h` (`OK`)	Rejected with `E7` `{studio_err}` (Studio 1 lacks drum equipment per `FR-14`).
TC-BB-06	R6	`lessonId="L-107"`, `instrument=DRUMS`, `teacherId="T-06"`, `quals=[DRUMS]`, `avail=true`, `contHours=3.5h`, `weeklyHours=39.0h`, `studioId="S-05"`, `studioState=FREE_DRUM_EQUIPPED`	`39.5h` (`OK < 40.0h`)	Confirmed with `E1` `{confirm_ok}`; teacher notified (`FR-15`) and audit logged (`FR-18`).
TC-BB-07A	R7	`lessonId="L-108"`, `instrument=PIANO`, `teacherId="T-07"`, `quals=[PIANO]`, `avail=true`, `contHours=2.0h`, `weeklyHours=39.5h`, `studioId="S-02"`, `studioState=FREE_PIANO_EQUIPPED`	`40.0h` (`WARNING`)	Triggers `E2` `{confirm_warn}` at exact `40.0h` warning boundary; confirms after manager override.
TC-BB-07B	R7	`lessonId="L-109"`, `instrument=PIANO`, `teacherId="T-07"`, `quals=[PIANO]`, `avail=true`, `contHours=2.5h`, `weeklyHours=43.5h`, `studioId="S-02"`, `studioState=FREE_PIANO_EQUIPPED`	`44.0h` (`WARNING`)	Triggers `E2` `{confirm_warn}` at exact `44.0h` upper warning boundary; confirms after manager override.

5 Wireframe

Because the Milestone 2 deliverable is submitted as a static PDF document, the user interface is presented through high-resolution static wireframe screenshots exported from Figma covering all required screen states, design system tokens, and functional workflows across the finalized requirements (FR-01 to FR-18 and NFR-USE-01 to NFR-COM-02).

5.1 Manager Scheduling & Outlier Intelligence View

- Linked Use Case: UC-01 (View Workload Dashboard & Outliers), UC-16 (Generate Studio Utilisation Report)
- Supported Requirements: FR-01, FR-02, FR-03, FR-15, FR-16, FR-17, NFR-USE-02.
- Screen Purpose: Serves as the operational landing cockpit for school managers, summarizing monthly faculty work allocations, assigned studio locations, studio utilisation metrics, and the 3-tier weekly workload indicators.
- Key Static Screen Elements Shown:
    1. 3-Tier Workload Status Indicators: Displays capacity meters and dual-encoded badges for the three workload tiers defined in FR-02, FR-17, and NFR-USE-02 (Green `OK < 40.0h`, Amber `Warning 40.0h–44.0h`, Red `Overworked > 44.0h`).
    2. Top 3 Lowest-Allocated Teachers: Highlights the 3 teachers with the lowest allocated hours at the top of the panel so managers can prioritize them during slot assignment.
    3. Studio Utilisation & Notification Controls: Displays booked versus available studio hours across the 86 weekly operating hours (FR-16) alongside confirmation controls that notify assigned teachers (FR-15).

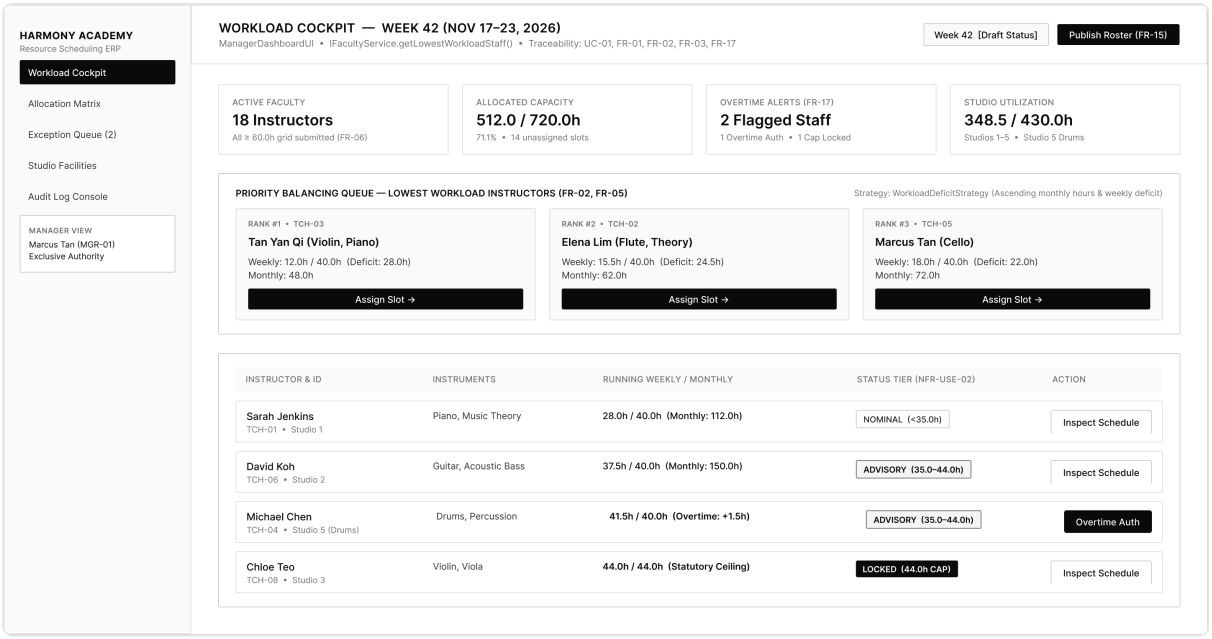


Figure 5.1: Manager Scheduling, 3-Tier Workload Status & Studio Utilisation Cockpit Wireframe (Static Screenshot — UC-01, UC-16, FR-01, FR-02, FR-03, FR-15, FR-16, FR-17)

5.2 Allocation Matrix & Conflict Feedback View

- Linked Use Case: `UC-02` (Allocate Lesson Slot), `UC-03` (Review & Re-allocate), `UC-06` (Substitute Reassignment).
- Supported Requirements: `FR-03`, `FR-04`, `FR-05`, `FR-11`, `FR-12`, `FR-14`, `FR-17`, `NFR-PER-02`.
- Screen Purpose: Provides managers with side-by-side candidate comparison when assigning a 30-minute lesson slot, displaying each candidate's current workload tier, teaching preferences, and predetermined studio location.
- Key Static Screen Elements Shown:
    1. Side-by-Side Candidate Panel: Displays up to 3 eligible candidates side-by-side showing current weekly/monthly workload, day/block preferences, and the assigned studio location for the selected date (`FR-04`).
    2. Tie-Breaking Strategy Ranking: Displays the active `ITieBreakerStrategy` ranking order (remaining capacity below 40.0 hours or preferred teaching day match per `FR-05`) with a `"Recommended"` badge.
    3. Studio Equipment & Maintenance Filtering: Displays studio equipment compatibility (`FR-14`, e.g., Studio 5 equipped with drums) and excludes studios scheduled for maintenance periods (`FR-11`).
    4. Warning Tier Overtime Prompt (`40.0h–44.0h`): Shows the overtime warning banner triggered when a slot places a teacher into the `40.0h–44.0h` warning tier (`FR-17`).

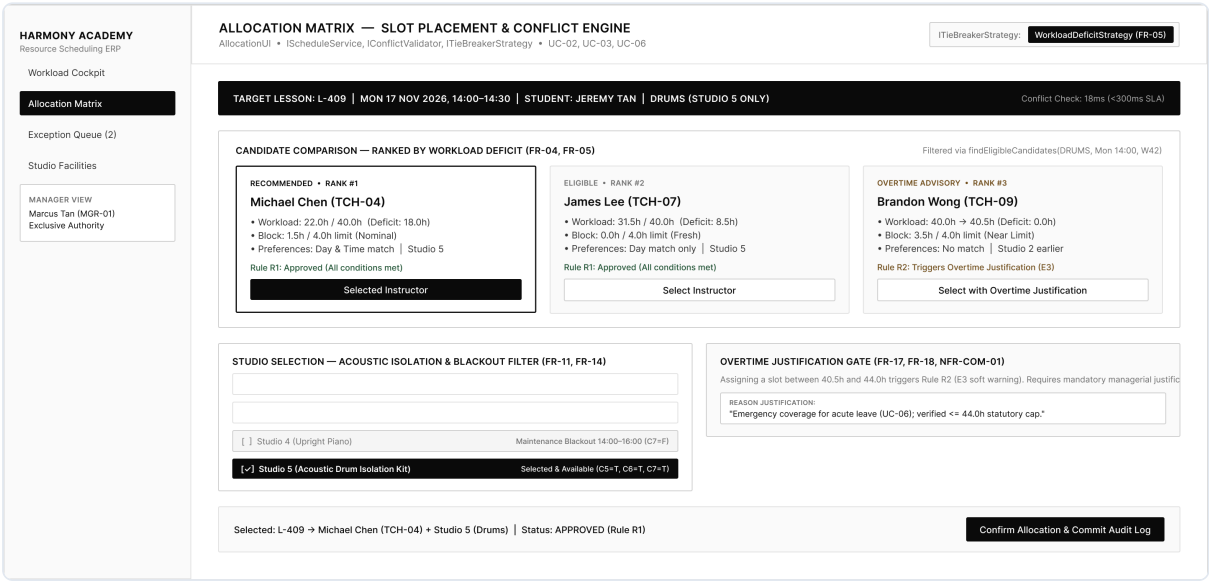


Figure 5.2: Allocation Matrix, Candidate Studio Location Display & Conflict Feedback Wireframe (Static Screenshot — UC-02, UC-03, UC-06, FR-04, FR-05, FR-11, FR-14, FR-17)

5.3 Teacher Availability & Timetable Inspection View

- Linked Use Case: UC-07 (Submit Availability), UC-08 (Submit Preferences), UC-09 (View Schedule), UC-10 (Reject Assigned Job).
- Supported Requirements: FR-01, FR-06, FR-07, FR-08, FR-09, FR-10, NFR-USE-01, NFR-SEC-01.
- Screen Purpose: Provides teachers with a portal to declare recurring monthly availability (minimum 60.0 hours), specify consecutive block preferences, view their confirmed weekly schedule with assigned studio locations and enrolled student details, and flag assigned jobs for rejection discussion with the manager.
- Key Static Screen Elements Shown:
    1. Recurring 30-Minute Availability Grid: Displays the monthly availability selector enforcing the 60.0-hour minimum declaration before the 19th 23:59 monthly cutoff (FR-06) and flagging post-19th edits for manager review (FR-10).
    2. Personal Weekly Schedule & Assigned Location: Displays confirmed lessons, assigned studio locations (FR-01), consecutive block progress against the 4.0-hour limit (FR-07), and enrolled student details (FR-08).
    3. Job Rejection Discussion Prompt: Captures a teacher's rejection request and routes it for discussion with the manager prior to re-allocation (FR-09).

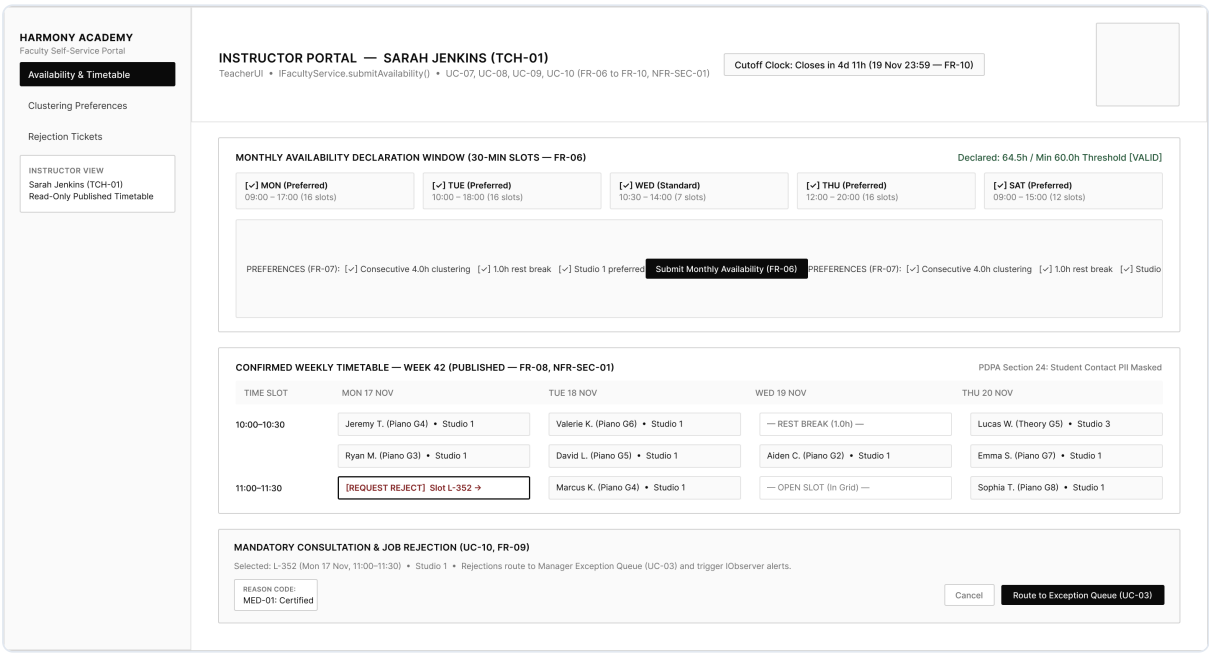


Figure 5.3: Teacher Availability Declaration, Personal Timetable & Job Rejection Wireframe (Static Screenshot — UC-07 to UC-10, FR-06 to FR-10)

5.4 Audit Log Console, Admin Configuration & Design System View

- Linked Use Case: UC-11 (Inspect System Audit Log), UC-12 to UC-15 (Account & Studio Maintenance), UC-17 (Review Labor Compliance).
- Supported Requirements: FR-11, FR-13, FR-14, FR-18, NFR-REL-02, NFR-COM-01, NFR-COM-02.
- Screen Purpose: Furnishes school managers and administrators with an immutable chronological audit trail of all lesson allocations, overtime confirmations, rejection updates, and studio/account configurations, alongside the standardized UI component token sheet.
- Key Static Screen Elements Shown:
    1. Multi-Parameter Audit Trail Filter: Enables querying by date range, actor ID, or action type (ALLOCATION_CONFIRMED, OVERTIME_WARNING_APPROVED, REJECTION_DISCUSSED, STUDIO_MAINTENANCE_SET) (FR-18).
    2. Regulatory Retention & Integrity Records: Displays immutable log entries retained for 24 months under Ministry of Manpower compliance rules (NFR-COM-01, NFR-COM-02).
    3. Unified Design System & 3-Tier Status Tokens: Showcases the reusable typography, spacing, studio badge, and 3-tier workload badge components (< 40.0h OK, 40.0h–44.0h Warning, > 44.0h Overworked) shared across all views.

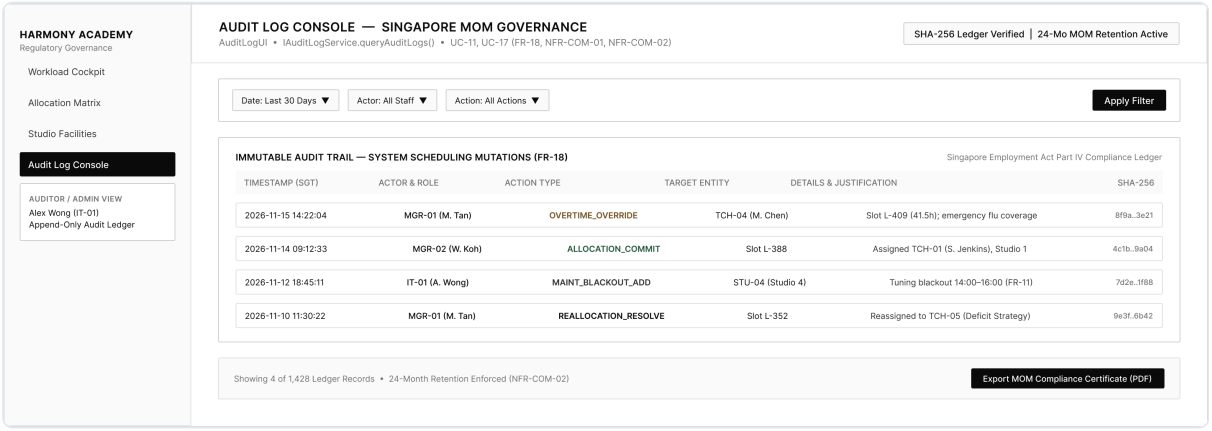


Figure 5.4: Regulatory Compliance & Immutable Audit Trail Console Wireframe (Static Screenshot — UC-11, UC-17, FR-18, NFR-COM-01, NFR-COM-02)

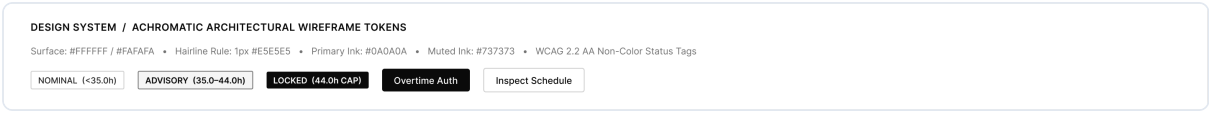


Figure 5.5: UI Design System Component Library & 3-Tier Workload Status Tokens (Static Screenshot — NFR-USE-01, NFR-USE-02)

6 References

1. Booch, G., Rumbaugh, J., & Jacobson, I. (2005). *The Unified Modeling Language User Guide* (2nd ed.). Addison-Wesley Professional.
2. Gamma, E., Helm, R., Johnson, R., & Vlissides, J. (1994). *Design Patterns: Elements of Reusable Object-Oriented Software*. Addison-Wesley.
3. McCabe, T. J. (1976). A Complexity Measure. *IEEE Transactions on Software Engineering*, SE-2(4), 308–320.
4. Ministry of Manpower Singapore. (2026). *Employment Act (Chapter 91) — Part IV: Hours of Work, Rest Days and Other Conditions of Service*. Singapore Statutes Online.
5. Personal Data Protection Commission Singapore. (2024). *Advisory Guidelines on Key Concepts in the Personal Data Protection Act (PDPA Section 24: Protection Obligation)*.
6. Pressman, R. S., & Maxim, B. R. (2020). *Software Engineering: A Practitioner's Approach* (9th ed.). McGraw-Hill Education.
7. World Wide Web Consortium (W3C). (2023). *Web Content Accessibility Guidelines (WCAG) 2.2 — Rule 1.4.1: Use of Color*. W3C Recommendation.